

MEMORY.md — Key Facts & Decisions

Core Concept

A daily 360° puzzle game. Player stands inside a geodesic sphere showing a mosaic (triangulated, darkened) version of a scene. They pick a corrupted tile, the view flips to an outside globe showing scrambled photo tiles, they find the match, and the real image snaps into place inside. Narrative: "the simulation is glitching — piece reality back together."

The Gameplay Loop — Mosaic Mode (SETTLED)

1. Player sees inside world: mosaic tiles in correct positions (abstract triangulated version of good-day image)
2. Long-press a mosaic tile → tile pulses amber for 900ms
3. View flips to outside: floating globe showing good-day photo tiles in WRONG (scrambled) positions
4. Preview thumbnail (bottom-left) shows the good-day content to find (flipped horizontally to match outside perspective)
5. Player swipes around globe to find matching photo tile
6. Long-press correct tile → green pulse on outside tile (650ms)
7. View flips back to inside, camera centers on placed tile, green pulse (800ms)
8. Placed tile now shows crisp photograph; source tile on outside gets swapped content
9. Repeat until all tiles transformed
10. Last remaining tile auto-completes

Two Game Modes (SETTLED)

- **Mosaic Mode** (default): Single image only. Bad day generated from good day via triangle mosaic + darkening. Inside/outside mechanic.
- **Image Mode**: Two Skybox AI panoramas (dystopian/utopian). Single-view pick-up/place mechanic. Original behaviour preserved.

Critical Technical Decisions (SETTLED)

Geodesic Dual Polyhedron

- Sphere built FROM hex+pentagon tiles
- Icosahedron → subdivide → project to sphere → compute dual
- Uniform hexagons + exactly 12 pentagons
- Subdivision 0=12, 1=42, 2=162, 3=642

Mosaic Texture Generator

- Samples good-day image at triangle centroids
- Draws solid-colour equilateral triangles (no edge lines)
- Darkened to 55% brightness for “corrupted” feel
- Detail slider: 2–200 triangles across width, default 98
- Canvas-based, generates a THREE.CanvasTexture

Inside/Outside View Mechanic

- Inside (default): camera at (0,0,0), material BackSide, looking outward. Shows mosaic (untransformed) or photo (transformed) tiles with CORRECT UVs
- Outside: camera at distance 780, material FrontSide, orbiting sphere. Shows photo tiles with SCRAMBLED UVs
- Automatic switching during gameplay: pick tile → outside, correct match → inside
- Manual toggle still available via settings panel
- Swipe controls: horizontal direction reverses between views, vertical stays same
- Momentum killed on every view switch

View Transition Feedback

- Amber pulse (900ms) on selected inside tile before switching to outside
- Green pulse (650ms) on matched outside tile before switching back
- Green pulse (800ms) on revealed inside tile after switch
- Camera centers on placed tile with momentum killed

- TRANSITIONING phase blocks input during animations

Scramble Map & UV Offsetting

- $sMap[i] = j$ means “outside position i holds good-day content for position j”
- Scrambled UVs computed per tile: UV offset from position i to $sMap[i]$
- On correct placement: $sMap[matched] = sMap[picked]$, UVs recomputed
- $findOutsideTile(p)$: reverse lookup — which outside tile holds content for position p

Preview Thumbnail

- Render-to-texture: Three.js renders target tile with goodTex from inside view
- Flipped horizontally in mosaic mode to match outside perspective
- Camera FOV calculated dynamically: $tileAngle * 1.8$
- Always renders from inside (BackSide) regardless of current view

Orange Tile Highlight (material tinting)

- Selected tile's material color tinted pulsing orange directly
- WebGL line width is always 1px — line-based highlights don't work

Render-to-texture for previews

- DO NOT go back to manual UV sampling or hex clipping

Embedded default images

- Ink Drawing panorama = bad day (base64 embedded)
- Dutch Masters panorama = good day (base64 embedded)
- Upload buttons available to swap images

Simple scramble swap (no chain resolution)

- On correct placement: $sMap[source] = sMap[target]$
- One move = one tile transformed
- Last tile auto-completes

Pole tiles

- Top tile = black with "OUT" text, Bottom tile = black with "IN" text
- Excluded from scramble and gameplay
- Reserved for future use

Three.js r128 + Capacitor

- Three.js for rendering, Capacitor for native wrapping

UV Mapping

- V coordinate: $v = \text{lat} / \text{Math.PI} + 0.5$
- X coordinate negated for inside-sphere view
- UV wrapping fix at 0°/360° seam
- RepeatWrapping on S axis, ClampToEdge on T
- Face internally subdivided for smooth texture interpolation

Input

- 400ms long-press threshold, 10px movement cancels
- Horizontal swipe reverses between inside/outside views
- Vertical swipe same in both views

Derangement

- Retry-until-valid shuffle, only for playable tiles

Grid lines

- 5 passes at slightly different radii for boldness
- Hex edges: cyan/blue, Pentagon edges: gold
- Off by default

Skybox AI Scenes

- Matched pairs confirmed via Photoshop overlay
- Equirectangular format (2:1 ratio)
- Only needed for Image Mode; Mosaic Mode uses single image

Game Features (SETTLED)

- Two-level hints (area cluster → exact tile, with cooldowns and score penalties)
- Easy/Hard mode toggle (preview thumbnail visibility)
- Inside/Outside view toggle (auto-switches in mosaic mode)
- Score, timer, progress bar, move counter
- Start/completion screens, basic Web Audio sounds
- Tile count slider (10–640 playable tiles)
- Detail slider for mosaic density (2–200)
- Mode selection: Mosaic or Image

Design Decisions (OPEN)

- Game name — “Better Day” is working title
- Narrative — “glitching simulation” direction emerging
- Pentagon special role
- Sound design investment
- Ambient soundtrack
- Marketing/promotion strategy
- Monetisation pricing
- Smooth animated view transitions (currently instant)
- Dimming matched tiles on outside view
- IN/OUT pole button functionality
- Developer control panel parameters

Approaches That Failed (DON'T REVISIT)

1. Flat hex grid on equirectangular — pole distortion
2. $\cos(\text{lat})$ correction — can't fix shape
3. Cubemap hex grids — seam artifacts

4. Spherical Voronoi — cells too irregular
5. Shader-based hex grid — didn't compile on iPhone
6. Manual UV sampling for previews — wrong content
7. UV offset/swap for scrambled display — seam bugs
8. Foreign tile content on another tile's mesh — UV mismatch
9. 2D overlay tracking 3D position — geometry misalignment
10. Auto-revealing neighbor tiles — caused stuck states
11. Chain resolution — unnecessary complexity
12. WebGL LineSegments for thick lines — always 1px
13. Hex clipping on preview render — misaligned crop
14. Mini-hexagon mosaic — hexagons don't subdivide evenly into hexagons, looks messy